SOP: Receipt Creation, Line Management & SQL Validation

System: Speed WMS
Sensitivity: Internal

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1. Purpose

This SOP describes the end-to-end process for creating and validating receipts in Speed WMS, including:

- **Receipt header creation**
- **Receipt line creation**
- **Stock generation behavior**
- **SQL validation and troubleshooting**

Applies to:
JetPark, Durban Northbound, and all Speed WMS warehouses.

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2. Core Database Tables (RAG Anchor)

Speed WMS receipt processing relies on the following tables:

- **REE_DAT → Receipt Header**
- **REL_DAT → Receipt Lines**
- **STK_DAT → Stock**
- **MVT_DAT → Movements**

These tables are referenced throughout this SOP.

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3. Receipt Header Creation

3.1 Objective

The receipt header represents the incoming goods transaction and acts as the parent container for receipt lines.

3.2 Access

A receipt header can be created using:

- **Add / New Receipt from the contextual menu, or**
- **The Add button on the receipt screen**

3.3 Receipt Header Structure

The receipt screen is divided into two sections:

- **Upper Section → Receipt header information (REE_DAT)**
- **Lower Section → Receipt lines (REL_DAT)**

3.4 Automatically Initialized Fields

The following fields are auto-populated and not editable:

- **Activity Code (ACT_CODE)**
- **Speed Receipt Number (REE_NORE)**
- **Receipt creation date and time**
- **Default reception dock (based on configuration)**

3.5 Mandatory User Inputs

The user must complete the following fields:

- **Customer Account (REE_CCLI) – if managed by the activity**
- **Supplier (Third Party Code)**
- **Receipt Reference (optional but recommended for traceability and search)**

3.6 SQL Validation – Receipt Header

Use this query to validate receipt header creation:

```
SELECT
  REE_KEYU,
  REE_NORE,
  REE_ETRE,
  REE_CCLI,
```

REE_TOP1

FROM REE_DAT

WHERE ACT_CODE = 'Activity Code'

AND REE_NORE = 'Receipt Number';

Key fields explained:

- **REE_ETRE → Receipt status**
- **REE_KEYU → Internal receipt identifier**
- **REE_TOP1 → Pegasus flag (1 = sent, 0 = not sent)**

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4. Receipt Line Creation

4.1 Objective

Receipt lines represent individual storage media that will later be converted into stock.

4.2 Minimum Required Information

Each receipt line must contain:

- **Item Code**
- **Quantity**
- **Quality Code**

4.3 Global Business Rules

- **Item code must exist in the article master**
- **Quantity is stored per storage medium**
- **Default quality is sourced from:**
  - **Item sheet, or**
  - **Activity default**
- **Traceability fields (Lot, DLC, DLUO, etc.) are mandatory if configured**
- **Owner priority follows this order:**

1. **Item owner**
2. **Activity owner**
3. **Manual owner selection**

4.4 Methods for Adding Receipt Lines

A. Manual Entry

- **Enter item code, quantity, and quality**
- **Validate to create the line**

B. Item Search

- **Right-click → *Search for Article***
- **Select item from:**
    - **Supplier list, or**
    - **All suppliers**
- **Enter quantity and quality**

C. Item Templates (Grid Entry)

- **Right-click → *Item Templates***
- **Select template**
- **Enter:**
    - **Same quantity for all items, or**
    - **Individual quantities**
- **Validate to create multiple lines**

D. Detailed Entry Form

- **Double-click item code field**
- **Enter logistic variants and traceability fields**
- **Validate to create the line**

E. Excel Import
Excel file must contain exactly three columns:

1. **Item Code**

2. **Quantity**
3. **Quality Code**

Steps:

- **Preview the import**
- **Validate to create receipt lines**

4.5 Receipt Line Status (Before Stock Generation)

Before stock generation:

- **Receipt lines exist in REL_DAT**
- **No stock exists yet**
- **Stock Input flag = 0**
- **No movement exists in MVT_DAT**
- **Lines can still be modified or deleted safely**

4.6 SQL Validation – Receipt Lines

```
SELECT
  REE_KEYU,
  REL_KEYU,
  REL_NOSU
FROM REL_DAT
WHERE ACT_CODE = 'Activity Code'
  AND REE_NORE = 'Receipt Number';
```

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5. Stock Generation Logic

5.1 What Happens During Stock Generation

When stock generation is executed:

- **Stock records are created in STK_DAT**
- **Receipt movements are recorded in MVT_DAT**

- **Expected receipt lines are consumed from REL_DAT**
- **Receipt status is updated in REE_DAT**

5.2 SQL Validation – Stock and Movements

Stock validation:

```
SELECT *
FROM STK_DAT
WHERE ACT_CODE = 'Activity Code'
 AND STK_NOSU = 'Support Number';
```

Movement validation (Receipt Movement):

```
SELECT *
FROM MVT_DAT
WHERE ACT_CODE = 'Activity Code'
 AND TMV_CODE = '10010';
```

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6. Reversal and Correction Scenarios

6.1 Reopening a Closed GRN

Use this SQL to reopen a closed receipt and reset Pegasus integration:

```
UPDATE REE_DAT
SET REE_ETRE = '100',
   REE_TOP1 = 0
WHERE ACT_CODE = 'Activity Code'
 AND REE_NORE = 'Receipt Number'
 AND REE_KEYU = 'Receipt ID'
 AND REE_CCLI = 'Customer Account Code';
```

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7. Troubleshooting Mapping (RAG-Friendly)

- **Receipt exists but no stock**

    - **Cause: Stock generation not executed**
    - **Table to check: REL_DAT**
- **Stock exists but no weight**
    - **Cause: Stock attributes not populated**
    - **Table to check: STK_DAT**
- **Receipt not sent to Pegasus**
    - **Cause: Pegasus flag not reset**
    - **Field to check: REE_TOP1**